340061 – Introduction to Graphics and Engineering Design

Winter Semester, 2026

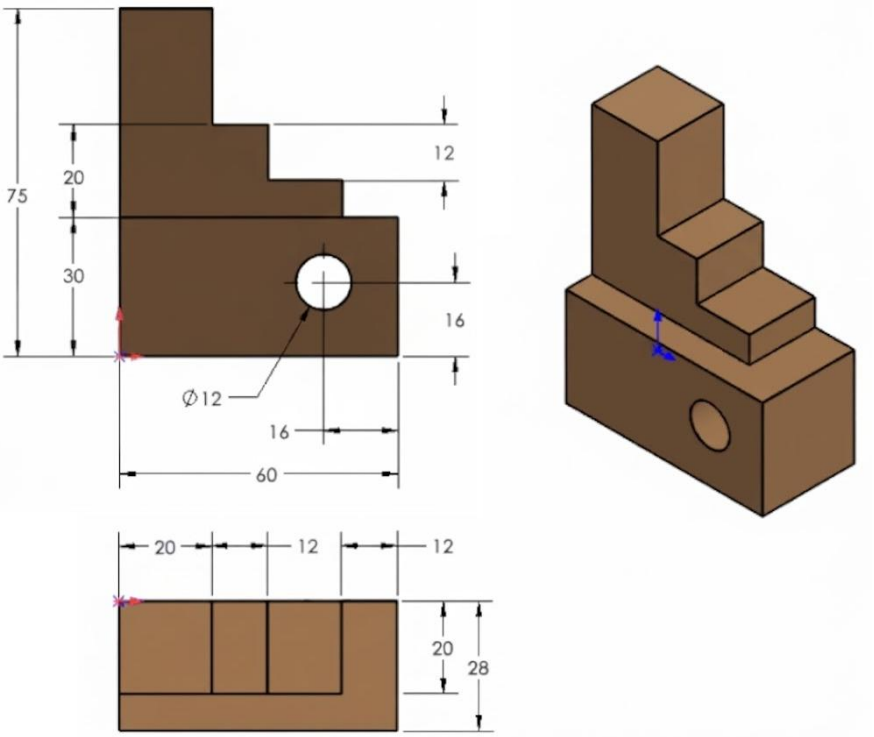
Lab 4
Part Modelling I

Instructions:

1. This document is aimed at self-practice during the lab. **No submission is required.**
2. Verify that all the sketches you create are **fully defined**.
3. All drawings are given in 1st angle projection. Verify that you define the front, top and right planes correctly.
4. The origin is given in each drawing in red. Design each part accordingly with the origin in the correct location.
5. Pay attention to the units - not all dimensions are in millimeters.

Q1.sldprt

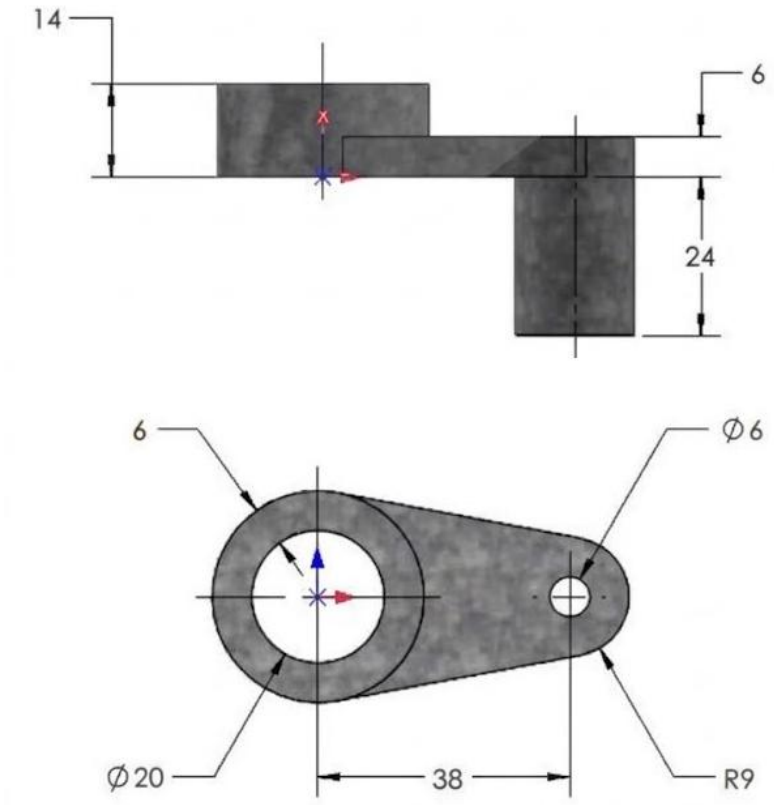
Model the following part:

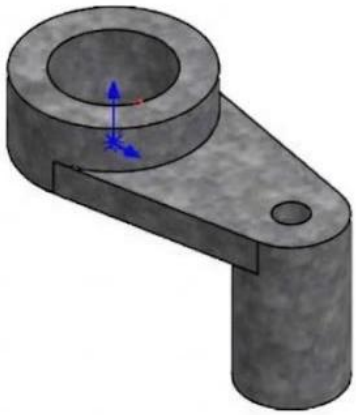


- The dimensions are in millimeters.
- Reference volume is $\mathbf{72593 \pm 1\%\ [mm^3]}$.
- For this part, use simple sketches for each feature.

Q2.sldprt

Model the following part:

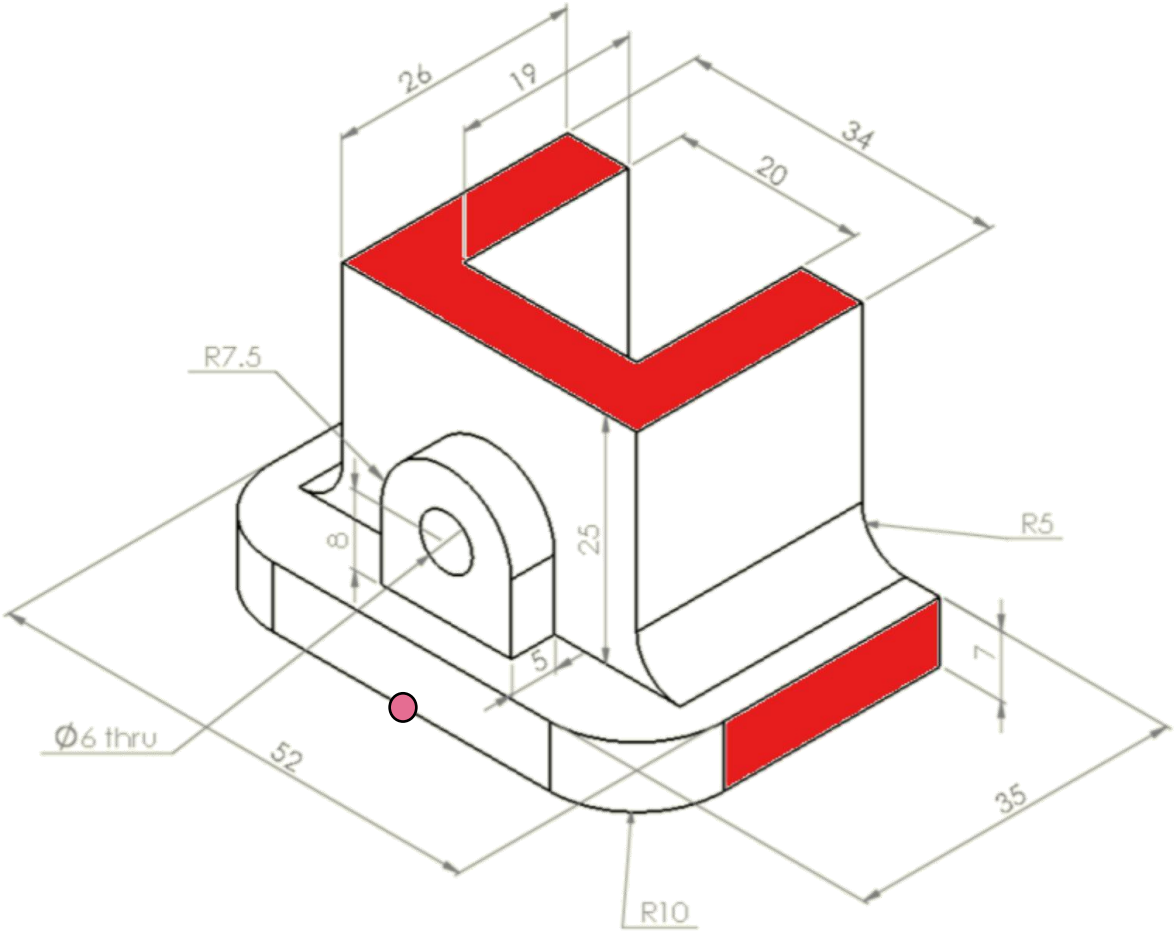




- The dimensions are in millimeters.
- Reference volume is $\mathbf{16268 \pm 1\%\ [mm^3]}$.
- Note that the symmetry plane is the **front plane**.

Q3.sldprt

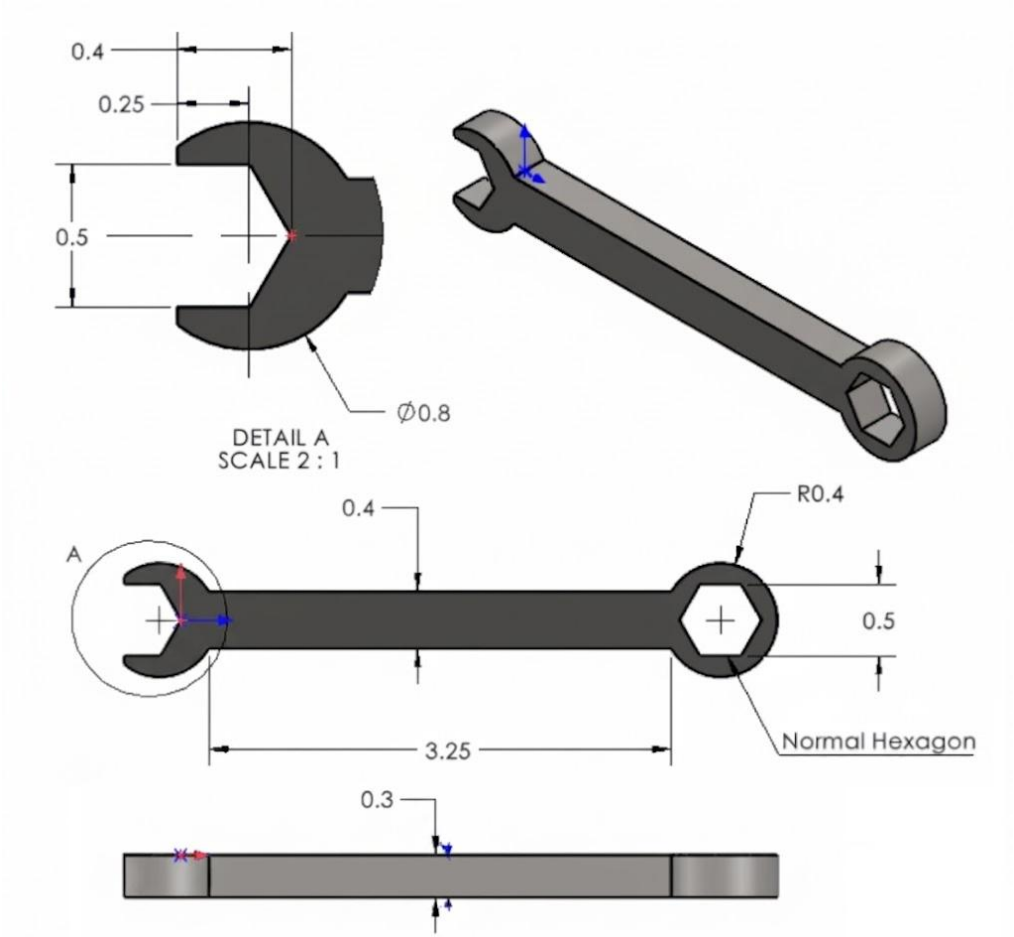
Model the following part:



- The dimensions are in millimeters.
- Reference volume is $\mathbf{23361 \pm 1\%\ [mm^3]}$.
- Design the part so that the symmetry plane is the **front plane**. Place the origin on the center bottom left edge (see pink dot).
- All cut holes are "Through all", including the 20mm opening at the rear.

Q4.sldprt

Model the following part:



- The dimensions are in millimeters.
- Reference volume is $\mathbf{0.5492 \pm 1\%\ [in^3]}$.
- Note that the symmetry plane is the **front plane**.

Q5.sldprt

Open Lab 3 file from Moodle and navigate to question 4. Look at the given front view, top view, and the isometric projection you drew, and use them to create a SolidWorks part.

Since exact dimensions are not given in the drawing, roughly approximate the measurements from the appearance in the drawing so that your model looks similar. Avoid measuring exact dimensions!

Design the part so that the symmetry plane is the front plane, as given in the drawing.

Do not use any features other than Extruded Boss and Extruded Cut.